

One AI platform behind three channels

Ecommerce, in-store point of sale and call-centre voice are three products. They are not three AI stacks. This is the platform that serves all three, taken apart layer by layer, with the build-or-buy call made explicit at every step.

CHANNELS	PLANES	DELIVERY	DATA RESIDENCY
3	7	6 phases · 12 months	AU by default

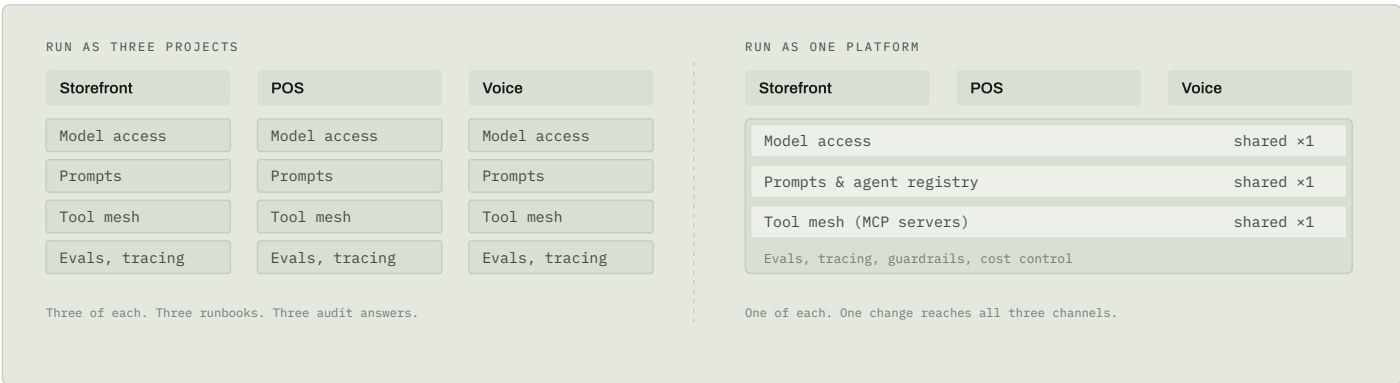
Written against the brief for an AI-first platform engineer at a multi-channel Australian adventure retailer: a Shopify replatform, a bespoke AI-first POS, and AI voice agents for the call centre. Grounded in a read of the retailer's public storefront, store directory, returns policy and support portal, so the integration inventory describes systems that actually exist rather than ones that might. Component names are generic; the decisions are not.

THESIS

Why a platform, and not three projects

Run the Shopify replatform, the POS build and the voice agents as three separate projects and they will each grow their own model integration, prompt store, tool layer, evaluation suite and audit trail. Build a capability three times and every change costs three times as much. Worse, every governance question gets three different answers, and none of them is defensible to an auditor.

The platform costs more in month one and far less by month nine. What makes it work is being strict about which capabilities are genuinely shared and which are genuinely channel-specific. Almost every failed "AI platform" is a shared layer that swallowed something it had no business owning.



The same four capabilities, three times over or once. Everything below the channel tabs is shared infrastructure. An improvement to retrieval, a cost control on the model gateway, or a new guardrail lands in all three channels at once, and there is a single audit trail to answer "what did the AI do, and why".

What is genuinely shared

Model access, routing and cost control. Tool access to systems of record, which is the expensive part. Retrieval over product, fitment and policy knowledge. Session and

customer memory. Guardrails, PII handling and audit. Evaluation, tracing and release gates. Prompt and agent versioning.

What is genuinely channel-specific

The transport: HTTP and server-sent events at one end, streaming audio at the other. The turn-taking model, because a caller expects an answer in under a second and a storefront shopper does not. The interaction contract, since a voice agent must say it is an AI and a staff copilot must not pretend to be the manager. And the fallback path, because a lost WAN link means something very different in a Campbelltown store than in a browser tab.

- RULE 1** The channel owns interaction.

Turn-taking, transport, disclosure and the degraded path live in the channel adapter, and nowhere else.
- RULE 2** The platform owns intelligence.

Models, tools, retrieval, memory, guardrails and evals are shared services with their own owners, versions and SLAs.
- RULE 3** The systems of record own truth.

The agent never becomes the source of a price, a stock level or a fitment. It reads them, cites them, and writes back through a governed path.

THE TELL

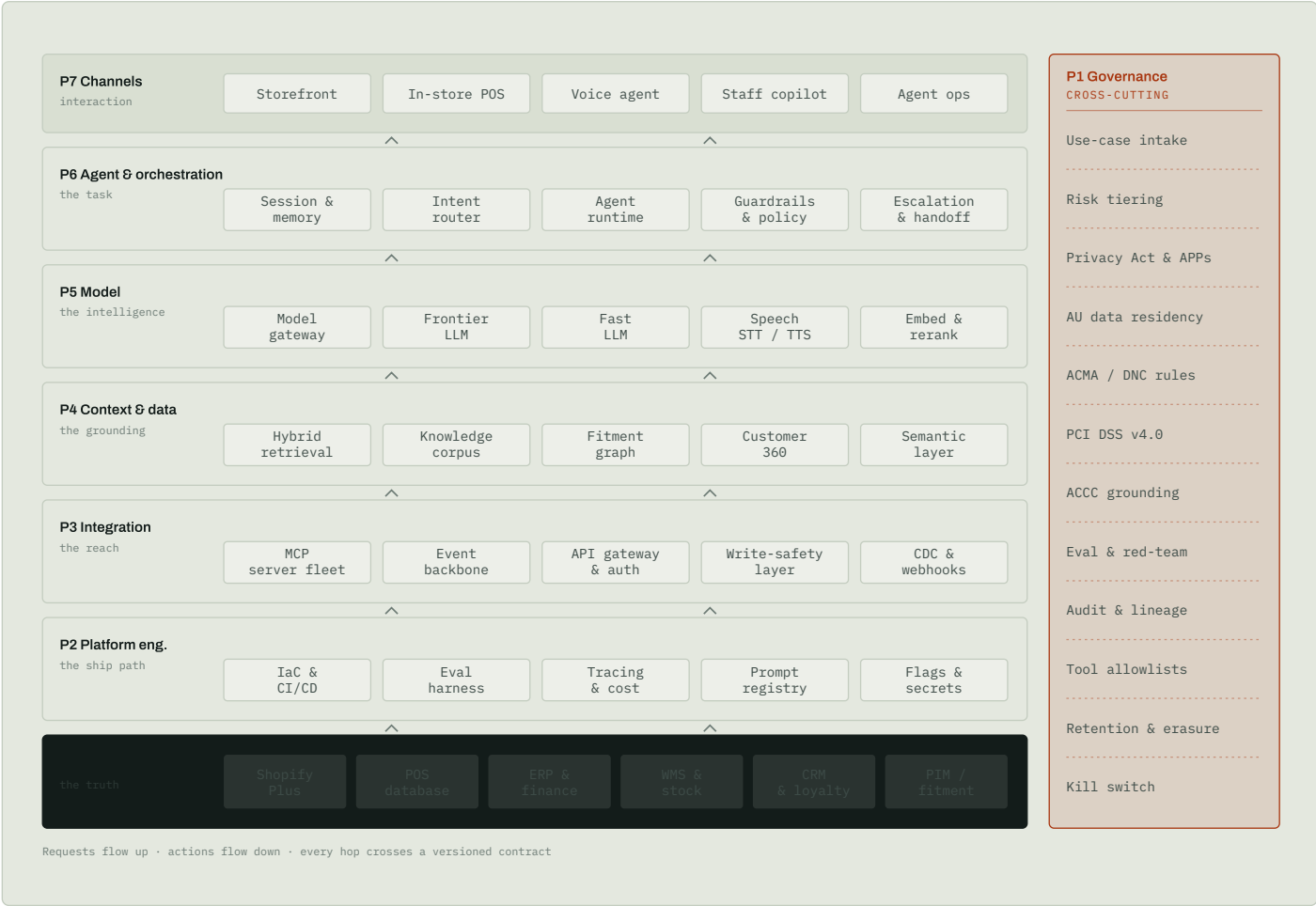
When a component starts making decisions in two planes at once, it is a bug. A prompt that formats a phone number for a voice agent has crossed into the channel. A policy rule hard-coded in the POS adapter has crossed out of governance.

ARCHITECTURE

Seven planes

Read the stack from the bottom up. Systems of record hold the truth. Plane 3 reaches into them through a governed tool mesh. Plane 4 assembles the context an answer needs. Plane 5 supplies the intelligence for one turn. Plane 6 turns that into a multi-step task with memory, guardrails and an escape hatch. Plane 7 speaks to a human in whatever medium they arrived in. Plane 1 runs down the side, because privacy, security, evaluation and audit are not a layer you pass through, they are constraints on every call in every other plane.

Requests flow up. Actions flow down. Every hop between planes crosses a versioned contract.



The seven planes. Plane 1 runs the full height because it constrains every call in every other plane rather than being a stage a request passes through. Note that the model planes (P5) are the thinnest and the least interesting: the architecture's weight sits in P3 reaching systems of record and P4 grounding answers in them.

COMPONENT INVENTORY

What each plane is made of

Each entry carries a build-or-buy call. The reasoning behind the calls is set out in Build or buy; the short version is that commodity intelligence gets bought, connective tissue and domain knowledge get built.

P7 · Channels

Thin adapters. Each one owns transport, turn-taking and the degraded path, and nothing else. Target: under 1,500 lines each.

Storefront assistant ADOPT + EXTEND Shopify theme app extension against the Storefront API. Falls back to native search when the agent is unavailable.	In-store POS copilot BUILD Staff-facing surface inside the bespoke POS. Runs against a local inference path so it survives a WAN outage.
Voice agent BUILD Telephony leg, streaming speech, barge-in, disclosure. The tightest latency budget in the system.	Staff copilot BUILD Internal surface in Slack or Teams. Doubles as the lowest-risk proving ground for the whole platform.
Agent ops console BUILD Escalation queue, transcript replay, live containment and cost, per-agent kill switch. Non-negotiable from day one.	

P6 · Agent & orchestration

The plane that turns a single model call into a task with memory, limits and an escape hatch.

Session serviceBUILD Channel-agnostic conversation state. A call, a chat and a POS session all resolve to one thread id, so context survives a handoff.	Intent routerBUY Fast model classifies the turn and decides: cached answer, direct response, tool call, or straight to a human. The main latency and cost lever.
Agent runtimeBUY Tool-calling loop with hard budgets: maximum tool calls, tokens and wall-clock per turn. Frameworks are commodity; budget enforcement is not.	Guardrail engineBUILD Input filters for injection and off-domain asks, output filters for groundedness, tone and invented values, and PII redaction on both sides.
Escalation & handoffBUILD Triggers on low confidence, sentiment, explicit request, complaint, legal or safety. Hands over a generated summary, not a cold transfer.	Agent & prompt registryBUILD Agents, prompts and tool bindings as versioned git artefacts with an owner, a changelog and a recorded eval score.

P5 · Model

One interface to every model, so no channel ever calls a provider SDK directly.

Model gatewayBUILD Policy-based routing on task, latency, cost and data class. Handles fallback, retries, caching, rate limits and per-cost-centre token accounting.	Frontier modelBUY Reserved for hard turns: ambiguous fitment, multi-step policy reasoning, complaint handling. Routed to an Australian region when the payload carries personal data.
Fast modelBUY Routing, classification, extraction, summarisation. Carries most traffic at a fraction of the cost and a third of the latency.	Speech stackBUY Streaming STT with partial hypotheses and streaming TTS optimised for first-byte time. Behind an abstraction, because this market moves fastest.
Embeddings & rerankerBUY Embedding model for the corpus index plus a cross-encoder reranker. Retrieval quality is mostly reranking, not embedding choice.	

P4 · Context & data

Where a generic assistant becomes this retailer's assistant. Most of the raw material already exists in some form; almost none of it exists in a shape an agent can use.

Retrieval serviceBUILD Hybrid lexical and vector search, metadata filters, rerank, and citation capture so every claim traces to a source document.	Knowledge corpusBUILD Roughly 87 articles are already published across Tips & Techniques, Complete Guides and DIY Projects, plus policy, freight and store operations material. It needs intent classification before it is usable, because destination content such as "Best Camping Spots Near Dandenong" is marketing rather than support and must never surface in a service answer.
Vehicle configuration modelBUILD Not a fitment matrix. The catalogue is universal-fit and named by dimension, so the asset is a model of the customer's vehicle and build: tub length and configuration, canopy or hard lid, roof load rating, existing 12V capacity, mounting interfaces. The hardest and most defensible thing in this architecture.	Policy rules engineBUILD Returns and warranty are rules, not prose: 30 days, store credit only, authorisation before return, exclusions for clearance and Scratch & Dent. Stored as date-versioned rules, because the published policy amends without notice and an agent quoting a superseded version is worse than one that defers.
Customer 360BUILD Identity resolution across POS, ecommerce, loyalty and CRM, so a caller is recognised whichever door they came through, with their wholesale tier and purchase channel attached.	Ingestion pipelineBUILD Connectors, change detection, freshness SLAs, ACL propagation, and a classification pass that tags every document by intent before it reaches an index.

P3 · Integration

Covered in full in the next section. This plane is where most AI programmes quietly fail.

MCP server fleetBUILD One typed server per system of record. The contract between agents and the business.	Event backboneBUY Order placed, stock moved, call started, cart abandoned. Feeds proactive behaviour and keeps context warm.
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API gateway & auth BUY One service identity per agent, scoped tokens, quotas and rate limits. No agent holds a human's credentials.	Write-safety layer BUILD Every agent-initiated write is idempotent, staged for human approval, or compensable. Read first, propose second, commit third.
CDC & webhooks BUILD Change data capture from systems of record so the context plane is minutes stale at worst, not overnight.	iPaaS connector BUY For long-tail internal systems that do not justify a bespoke server. A holding pen, not a destination.

P2 · Platform engineering

How the team ships, and the plane that makes AI changes safe to release on a Tuesday afternoon.

IaC & CI/CD BUILD Terraform for infrastructure, ephemeral preview environments per branch, standard pipelines for services and prompts alike.	Eval harness BUILD Golden datasets per agent, scored automatically on every pull request. A prompt cannot reach production without a passing eval delta.
Tracing & cost BUY Distributed traces across speech, router, tools, model and output, with token and dollar cost attributed per channel and per conversation.	Prompt registry BUILD Prompts as versioned artefacts with rollback, so a prompt change is a deployable unit with an audit trail rather than a dashboard edit.
Flags & rollout BUY Shadow mode lets a new agent listen alongside the live one before it is allowed to speak. Canary by store, by queue, by percentage.	Secrets & KMS BUY Managed keys, egress allowlists, and a network boundary that makes the model gateway the only path out.

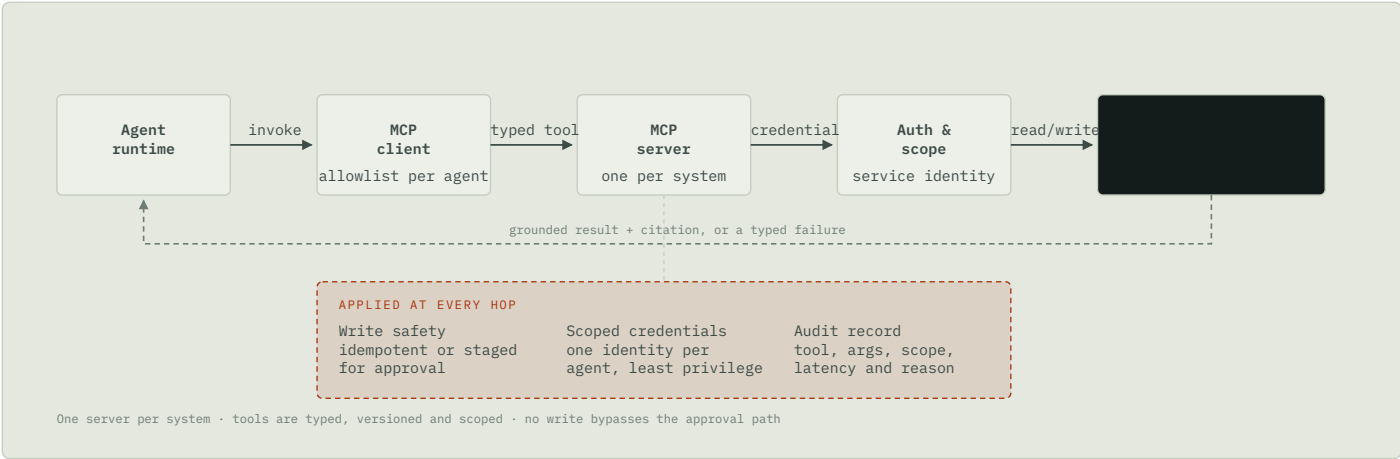
PLANE 3

The integration plane is the product

AI programmes rarely fail on models. They fail on plumbing. The model is a weekend's work; getting a governed, typed, auditable read and write into a fifteen-year-old ERP is a quarter.

So the integration plane gets treated as the platform's main deliverable. Every system of record exposes a Model Context Protocol server: a small, typed, versioned surface that declares exactly which operations an agent may perform, with what arguments, under which credential. Agents never speak REST. They never hold a database credential. They call tools.

The payoff is that integration becomes a reusable asset instead of glue buried in a prompt. A new agent gets the whole business on day one. A security review covers one surface rather than nine. And when the ERP is replaced, one server changes and no agent knows.



One tool call, end to end. The agent holds no credentials and knows no endpoints. It holds a list of permitted tool names, and the MCP server is the only thing that knows how to talk to the

system behind it. Writes take the same path but stop at the approval gate before the credential is minted.

SYSTEM OF RECORD	TOOLS IT EXPOSES	ACCESS	WHY THE AGENT NEEDS IT
Shopify Plus	order.get, order.status, cart.recover, product.search, return.request	read · gated write	Order status, post-purchase service, returns intake
POS database	sale.lookup, txn.history, refund.propose	read · propose only	In-store sale history and the approval position for a return
Store stock BUILD	stock.companyStore, stock.transferEta	read	Store-level stock is not exposed anywhere today. It has to exist before the agent can answer the most common question it will be asked
Stockist directory BUILD	stockist.locate, stockist.confirm	read	Roughly 144 third-party outlets with no stock visibility at all. The agent may only ever say "call to confirm"
Store directory BUILD	store.locate, store.hours, clickcollect.request	read · write	Per-store trading hours are not published per store, and Click & Collect is arranged by phone
ERP & finance	invoice.get, credit.status, supplier.eta	read	Trade accounts, backorder dates, freight cost
WMS & warehouse	inventory.bySku, backorder.eta, shipment.stage	read	National position. Distinct from store stock, and the two must never be conflated
CRM & loyalty	customer.get, interaction.log, tier.get, wholesale.eligibility	read · gated write	Wholesale pricing unlocks at four or more items, so eligibility changes what a quoted price should be
PIM / catalogue	product.get, variant.get, specs.get, stockclass.flag	read	Clearance, factory seconds and Scratch & Dent carry different return rights, so the class is part of the answer
Returns & warranty	returns.authorise, warranty.assess, credit.issue	gated write, human approves	No return is processed without approval, and faulty items are assessed before a claim is finalised
Payments	payment.link, payment.status	write, no PAN	Afterpay, Zip and PayPal are offered. Card data never enters a voice or chat context
Freight & carriers	shipment.track, delivery.book	read · gated write	"Where is my order", and booking a fitting or collection slot
Help centre	route.sales, route.tracking, route.returns, route.general	write	Escalations land in the four queues staff already work, instead of a fifth one nobody monitors

THE GAP THE SITE REVEALS

Store-level stock is not exposed anywhere. Today the only way to find out whether a Campbelltown store has an item is to phone 1800 88 39 64 and ask, and Click & Collect is arranged on that same call. That is the most common reason a customer rings, and the business cannot currently answer it in software. No agent fix for "have you got it at Campbelltown" can exist until that data does.

Alongside it sit roughly 144 third-party stockists, mostly hardware chains like Mitre 10, Home Hardware, True Value and Autopro, holding stock the business has no visibility into whatsoever. The agent must be able to say "call the store to confirm" without ever implying an availability it cannot verify. A wasted trip is the most expensive failure this system can produce.

CORRECTING AN EARLIER ASSUMPTION

The first version of this architecture called for a fitment graph: make, model, year and variant matched against part. The catalogue says otherwise. Products are named by size and capacity, not by vehicle: 900mm and 1300mm drawers, a 12,000lb winch, 20, 30 and 40 inch lightbars, and one roof rack sold as a "Universal Ute Tub Rack". No vehicle data appears anywhere, and bull bars and suspension are not listed at all. There is no fitment matrix to buy or build.

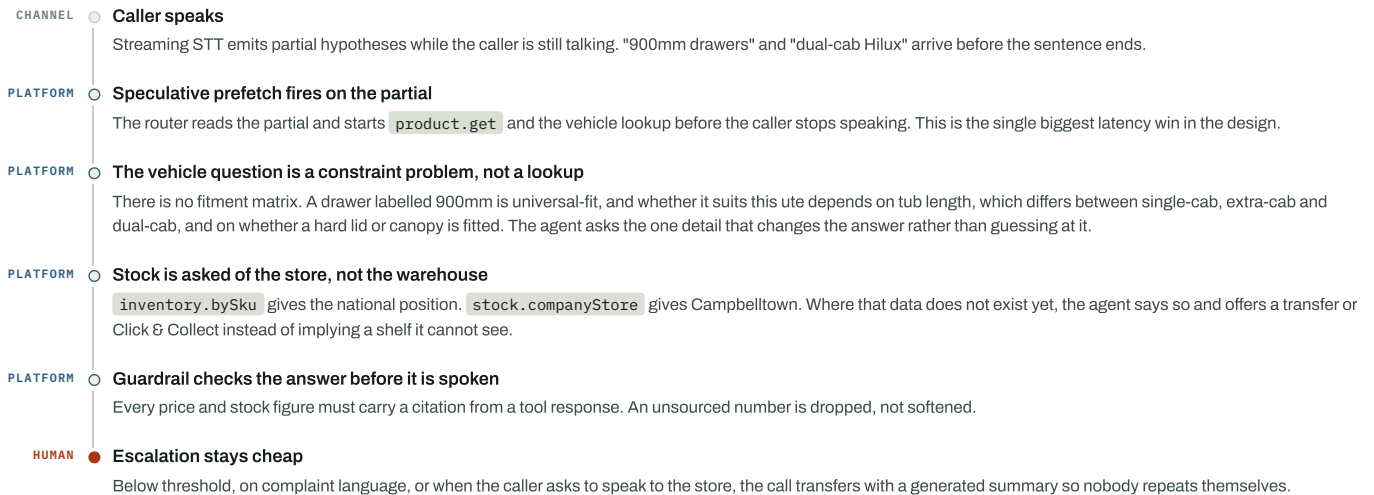
The question does not disappear, it changes shape. "Will this fit my ute" becomes a constraint problem: whether 900mm drawers suit this tub, which varies by cab configuration, and whether a hard lid or canopy is fitted. Whether a roof top tent's weight sits under this vehicle's roof load rating, and whether cross bars are needed. Whether a lightbar of this width fouls this nudge bar. That is harder to build than a lookup table, and correspondingly harder for anyone else to copy.

Three requests, traced

The same platform, three very different contracts. Watch where the human stays in the loop, and where the agent is allowed to commit.

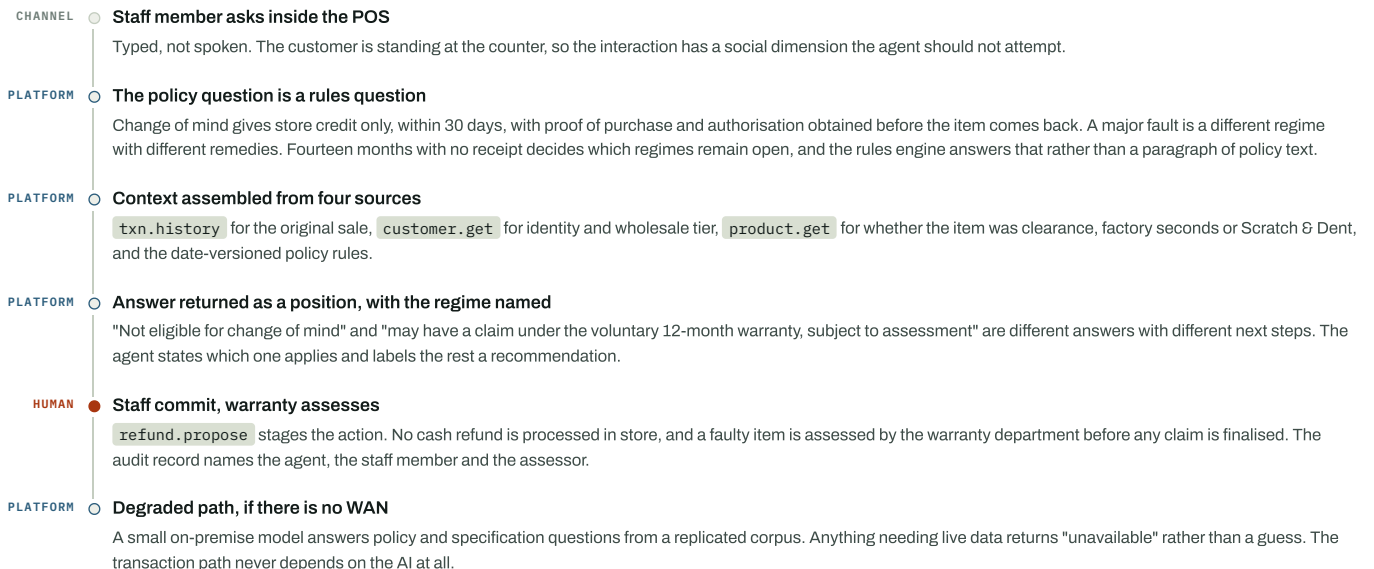
Call centre: "Will the 900mm drawers fit my dual-cab Hilux, and have you got them at Campbelltown?"

The hardest of the three. The caller expects an answer before they finish the question, and the two halves of that sentence need completely different systems.



In-store: "Customer wants a refund on a winch bought fourteen months ago, and the receipt is gone."

A policy question disguised as a product question, and the clearest argument in this document for keeping a human on the commit.



Storefront: "Will this roof top tent fit my Prado with factory roof rails?"

The only channel where the customer is alone, unsupervised and about to spend money.

- CHANNEL** ○ **Shopper asks on the product page**
The SKU being viewed is passed in automatically. The shopper should never have to name the product they are looking at.
- PLATFORM** ○ **Vehicle profile pulled from the session**
If the shopper has registered a vehicle it is reused. If not, the agent asks for the minimum it needs and offers to save it for next time.
- PLATFORM** ○ **The answer is a load calculation, not a compatibility flag**
The tent's weight against this vehicle's roof load rating, whether factory rails are present, and whether a cross bar kit is required. The reply names the tent, the cross bars it needs, and the rating it has to stay under.
- PLATFORM** ○ **Alternatives are offered, not withheld**
If the rating is insufficient the agent names what does fit, rather than leaving the shopper at a dead end with a "no".
- HUMAN** ● **Low-confidence answers route to a person**
With the vehicle, the SKU and the reason for uncertainty already attached to the enquiry, so the customer waits for an expert instead of receiving a plausible invention.

NON-FUNCTIONAL TARGETS

What good looks like, per channel

These are the numbers the platform is designed against. They are written as SLOs with an owner, and they are asserted in CI rather than discovered in production.

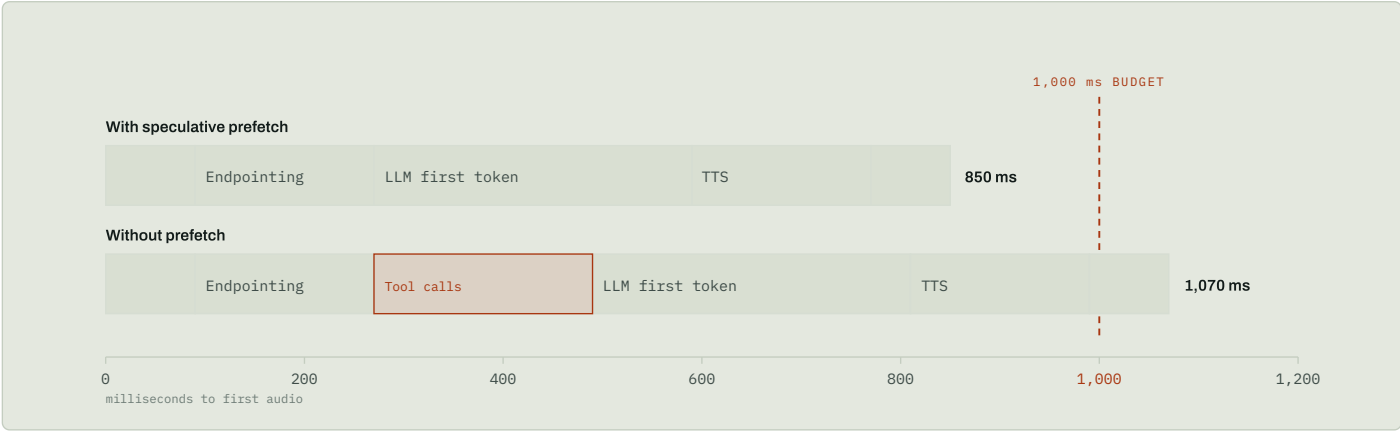
CHANNEL	LATENCY SLO	AVAILABILITY	DEGRADED PATH	QUALITY GATE
Voice	p50 < 850 ms · p95 < 1.2 s first audio	99.9%	Warm transfer to a human queue	Containment ≥ 40% · zero invented stock or price
In-store POS	assist < 1.5 s · POS UI < 300 ms	99.95% core POS	On-premise model, then a static policy lookup	Fitment accuracy ≥ 99% on the eval set
Storefront	p95 < 1.2 s first token	99.9%	Native Shopify search and help centre	Groundedness ≥ 98% · every claim cited
Staff copilot	p95 < 2 s	99.5%	Static knowledge base	Citation rate 100%

AVAILABILITY IS NOT ONE NUMBER

Core POS must clear 99.95% and the AI on top of it does not need to. If the assist layer is down, staff sell with the deterministic POS they have always had. Making the AI a dependency of the transaction path would be the single most damaging design mistake available here.

The voice latency budget

A caller tolerates roughly a second of silence before they assume the line has dropped. That second has to cover telephony, speech recognition, turn detection, retrieval, model reasoning, speech synthesis and playback. There is no way to spend it sequentially, so the design buys time by starting work before the caller stops speaking.



Where the second goes. The two bars are the same turn. The only difference is the orange block: tool calls issued against the partial transcript while the caller was still speaking. Everything else, telephony and jitter buffer, endpointing, model time to first token, synthesis and playback buffering, is unchanged. Speculative prefetch also needs a miss path, roughly 220 ms added, so the router only speculates when the partial already names a vehicle and a part category.

- 01

Speculate during speech.
Start tool calls on the partial transcript. Abandon them cheaply if the finished sentence contradicts the hypothesis.
- 02

Two model tiers on the critical path.
The fast model answers routine turns outright. Only hard turns escalate to the frontier model, and only after a filler has been played.
- 03

Endpointing beats silence timers.
Semantic turn detection removes roughly 400 ms of dead air compared with waiting for a fixed pause.
- 04

Keep connections warm.
Pre-established model and speech sessions. A cold TLS handshake at turn one is a second of silence by itself.
- 05

Barge-in is mandatory.
The caller must be able to interrupt. A voice agent that talks over people is worse than an IVR menu.

SOURCING

Build or buy

The brief asks for expertise on platforms and an ability to build where a platform falls short. The line runs in a consistent place: **buy the commodity intelligence, build the connective tissue and the domain knowledge**. Models, speech and vector search are improving faster than any internal team can, so owning them is a liability. The tool mesh, the vehicle configuration model and the eval suite are specific to this business, so renting them is a liability.

CAPABILITY	CALL	REASONING	EXIT COST IF WRONG
Frontier & fast models	buy	Commodity, improving monthly, no differentiation in owning it	low, behind the gateway
Model gateway & routing	build	Thin layer. It is the only thing preventing provider lock-in and uncontrolled spend	low
Speech STT / TTS	buy	Fastest-moving segment in the stack. Abstract it, do not own it	low, one adapter
Voice orchestration	buy, then fork	Realtime plumbing is solved; start on a platform, take ownership of the tool layer immediately	medium
Agent framework	buy	Tool-calling loops are commoditising. The budgets and policy around them are not	medium
MCP server fleet	build	The majority of the work and the source of all reusability	high if rented poorly
Retrieval & rerank	buy components	Managed vector search is fine. The corpus pipeline is yours	low

CAPABILITY	CALL	REASONING	EXIT COST IF WRONG
Vehicle configuration model	build	Products are universal-fit, so there is no fitment matrix to buy. Modelling the customer's vehicle and build is the differentiator	n/a, no alternative
Eval harness	build	A vendor does not know what a wrong fitment answer looks like here	high
Tracing & cost	buy	Solved problem. Buy the traces, build the channel dashboards	low
Shopify merchant AI	adopt	Use what ships natively for merchandising and merchant ops. Do not rebuild it	n/a
Bespoke POS AI	build	The POS is already differentiated and offline constraints rule out most platforms	n/a
Human handoff & desktop	adopt	Extend the existing contact centre rather than running a parallel one	medium

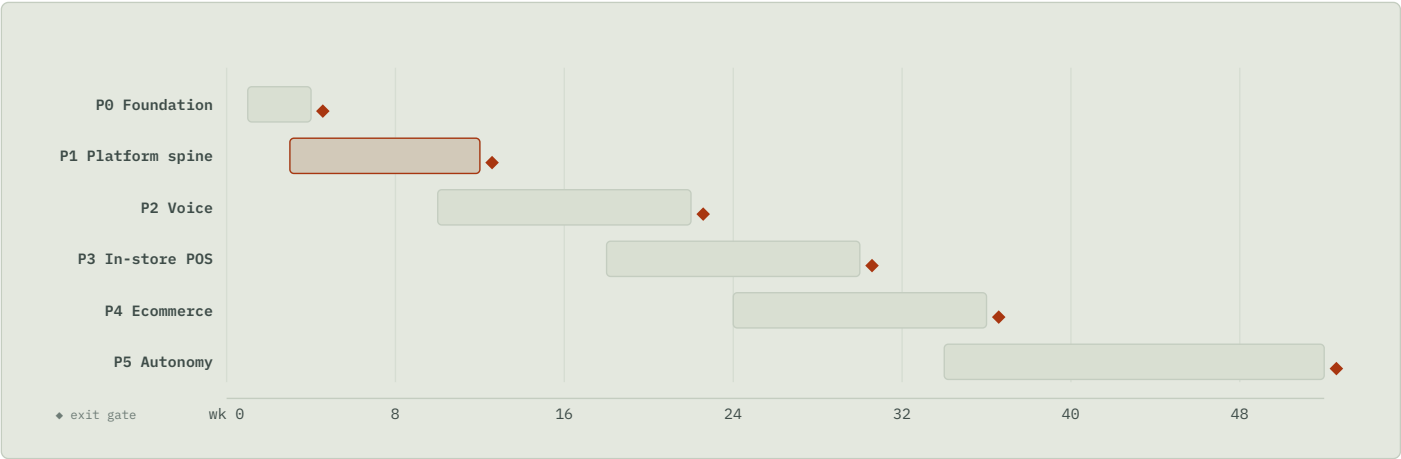
THE TRAP IN "BUY"

A bought platform is only safe if the thing you actually depend on stays yours. Buy the speech engine, but keep the tool mesh, the prompts and the eval suite inside the platform. Otherwise the day the vendor changes pricing or gets acquired, the migration is not a re-integration, it is a rebuild.

DELIVERY

Six phases, twelve months

Sequence matters more than speed here. Each phase leaves behind a platform capability that the next phase consumes, and the riskiest channel is deliberately not first. Voice is the highest-value and highest-risk build, so it lands after the platform spine has been proven on an audience that cannot generate a viral complaint.



Overlap is deliberate. Voice engineering starts at week 10, two weeks before the spine is declared done, because telephony procurement and carrier integration have long lead times. Phase 5 never really ends; it is the steady state the platform runs in once the three channels are live.

P0 Foundation

weeks 1-4

- SCOPE
- Inventory every AI use case, scored by value and risk tier
 - Map systems of record, data owners and integration constraints
 - Assess catalogue data quality and the vehicle attributes the agent will need, which determines what it can promise
 - Ratify the architecture decisions and the sourcing calls
 - Privilege, privacy and security baseline; residency position
 - Baseline today's numbers: call containment, handle time, return rate

EXIT GATE

Architecture signed, decision records published, risk register owned, baseline instrumented. No model is called before this gate.

P1 Platform spine

weeks 3–12

SCOPE

- Model gateway with routing, fallback and cost accounting
- MCP servers for the first five systems: Shopify, POS, inventory, CRM, policy
- Retrieval service, corpus pipeline with intent classification, and the vehicle configuration model
- Store and stockist directories, which unblock the most common question the agent will face
- Session, memory, guardrails and the eval harness
- Tracing, per-conversation cost, prompt registry, CI eval gates
- Proven on the lowest-risk audience: an internal staff copilot

EXIT GATE

One internal agent in daily staff use. Eval suite green in CI on every pull request. Cost per conversation measured and attributed. Tool mesh covers the five core systems.

P2 Voice agent, containment scope

weeks 10–22

SCOPE

- Telephony integration, streaming speech, barge-in, warm transfer
- Latency budget met end to end, asserted in CI against a recorded corpus
- AI disclosure, consent capture and Do Not Call Register washing
- Containment limited to order status, freight tracking, store location, and specification questions the published guides can ground
- Store stock enters containment only once that data exists, which may land later than the voice agent itself. Scoping it in early would promise a capability the business does not yet have
- Shadow mode first: the agent listens alongside the live queue before it speaks

EXIT GATE

First-audio p50 under 850 ms in production. Containment at or above 40%. Zero compliance incidents. Human-handled CSAT not degraded. Rollback exercised and proven.

P3 In-store POS

weeks 18–30

SCOPE

- On-premise inference path and the offline degradation ladder
- Staff copilot for policy, warranty, fitment and compatible upsell
- Customer-facing fitment confirmation at the counter
- All write actions staged for staff approval, with dual attribution in the audit record
- Pilot in a small store cohort before rollout

EXIT GATE

No POS downtime attributable to the AI across the pilot. Vehicle-fitment answers correct on the eval set, including the ones the agent correctly declines to answer. Measured staff adoption, not just licence deployment.

P4 Ecommerce and Shopify

weeks 24–36

SCOPE

- Storefront assistant on the product page, scoped to fitment and specification
- Post-purchase deflection: where is my order, warranty, returns intake
- Catalogue enrichment and attribute completion, human-reviewed
- Consent-based cart recovery, never an autonomous discount
- Native Shopify AI adopted for merchant-side operations

EXIT GATE

Deflection rate measured against the phase 0 baseline. Groundedness gate at 98%. Return rate and average order value effects reported, including the negative results.

P5 Autonomy and optimisation

weeks 34–52+

SCOPE

- Consent-based proactive outbound, within telemarketing rules
- Personalisation across channels off one customer profile
- Risk-tier promotion of specific actions from propose to commit
- Cost and latency optimisation, model migration without channel changes
- Governance forum operating on a standing cadence

EXIT GATE

Promotion process proven end to end, with a documented case of an action being promoted and another being demoted. Unit economics positive per channel.

PLANE 1

Governance, risk and compliance

Australia has unusually specific obligations for exactly the things this platform does. An AI voice agent that calls customers is telemarketing. A hallucinated price is a misrepresentation under consumer law. An AI that decides a refund is an automated decision with privacy consequences. These are design inputs, not a review at the end.

Privacy Act & the APPs Call recordings and transcripts are personal information. Cross-border disclosure rules apply the moment a prompt reaches an offshore endpoint, so residency routing is a control, not a preference. Notifiable breaches run through the OAIC.	Automated decision-making Recent reforms tighten obligations where a computer program makes a decision that significantly affects an individual. Refund approval and warranty denial sit squarely in scope. The design keeps humans on consequential decisions and logs the model's recommendation alongside the human's.
Telemarketing & AI disclosure Outbound AI calls are telemarketing. Consent must be captured, the Do Not Call Register must be washed, and the agent must disclose that it is an AI. Spoofing a human agent is prohibited and also a brand decision no one should want.	PCI DSS v4.0 The moment a card number enters a voice or chat context, that whole system inherits cardholder-data scope. The design never lets it happen: payment links, DTMF masking at the carrier, and transcript redaction as a hard output filter.
Consumer law grounding Misleading conduct applies to machine statements. An invented stock level or fitment claim is a misrepresentation whether or not a human typed it. Groundedness is therefore a legal control with an eval threshold, not a quality nicety.	Residency & vendor assessment Personal data and transcripts stay in an Australian region wherever the provider offers one. Where it does not, the disclosure is documented and assessed rather than assumed away.
Retention and erasure A retention schedule for transcripts, and a working erasure path that reaches the vector index. Embeddings of personal data are still personal data, and this is the step most teams discover they cannot perform.	Kill switch & audit lineage Per-agent and per-channel stop control with a tested runbook. Every call traceable to the model version, prompt version, tool versions and retrieved documents that produced it.

RISK TIERING DRIVES EVERYTHING ELSE
Every use case is classified before it is built: **L1** advises a human, **L2** acts but requires approval, **L3** acts autonomously within limits. The tier sets the eval threshold, the logging depth, the human-in-the-loop requirement and the blast radius. Nothing is promoted to L3 by enthusiasm; it is promoted by evidence from a shadow run.

DECISION RECORDS

The calls that shape everything else

Each of these is expensive to reverse. They are recorded with the trade-off that was accepted, because the alternatives are reasonable and will be raised again.

ADR-001

MCP servers are the only path from agent to system of record

CONTEXT

Nineteen systems, three channels, and a team of fifteen. Direct API calls from each agent would produce three parallel integrations per system and no shared contract.

DECISION

One typed MCP server per system, owned by the platform team, versioned and eval-covered like any other service.

TRADE-OFF

An extra hop and a schema to maintain. In exchange, no agent holds credentials, no agent knows an endpoint, and a system can be replaced without touching a prompt.

ADR-002

No channel calls a model provider directly

CONTEXT

Model prices, capabilities and regional availability change on a monthly cadence. Three channels calling three SDKs makes every change a three-way migration.

DECISION

All inference through one gateway that owns routing, fallback, caching, budget caps and per-cost-centre accounting.

TRADE-OFF

The gateway can lag a provider's newest feature by days, and becomes a shared dependency that must never be down. Accepted, because provider lock-in across three channels is the larger risk.

ADR-003

The POS keeps working with no network, and the AI is not on its critical path

CONTEXT

Stores lose connectivity. A cloud-only assistant turns a WAN outage into a queue at the counter.

DECISION

A small on-premise model answers policy and fitment from a replicated corpus. Anything needing live data returns unavailable rather than a guess. The transaction path never depends on inference at all.

TRADE-OFF

A second deployment target, a replication pipeline, and a capability that differs from the cloud path. Accepted, because a checkout that stops is unrecoverable and an assistant that degrades is not.

ADR - 004 Buy voice orchestration, build the tool layer

CONTEXT	Real-time audio transport, jitter handling and interruption logic are genuinely hard and genuinely solved by vendors. Tool integration into this business is neither solved nor for sale.
DECISION	Start on a voice platform for the realtime leg. Keep the tool mesh, prompts and evals in the platform from day one, behind an interface thin enough to swap the orchestration layer.
TRADE-OFF	A vendor sits on the critical path for the highest-value channel. Mitigated by keeping the replaceable part small and the owned part large.

ADR - 005 No prompt reaches production without a passing eval delta

CONTEXT	Prompts are edited in dashboards, by many people, with no review and no history. A single word change can alter refund behaviour across three channels.
DECISION	Prompts and agent configurations live in git beside the code they drive. CI scores every change against golden datasets and refuses regressions.
TRADE-OFF	Slower iteration and real work to build the datasets. Accepted, because unreviewed prompt changes are the most common way an AI system degrades without anyone noticing.

ADR - 006 Personal data is routed to Australian regions by default

CONTEXT	Transcripts, customer records and vehicle details flow through every inference call. Offshore endpoints are cheaper and sometimes faster.
DECISION	The gateway classifies the payload and routes anything carrying personal information to an Australian region, accepting a cost and capability penalty where one exists.
TRADE-OFF	Some models arrive later in Australian regions, and a few not at all. Accepted, because cross-border disclosure of call recordings is a compliance exposure that does not reduce over time.

ADR - 007 Humans commit every money-moving action

CONTEXT	Refunds, credits, discounts and warranty outcomes are exactly where an AI error is most expensive and most visible.
DECISION	Agents gather, reason and propose. A human commits. Both are named in the audit record. Promotion to autonomous commit requires a shadow run and an explicit risk-tier review.
TRADE-OFF	The labour saving is smaller than the demo implies. Accepted, because a single mis-issued refund to the wrong customer costs more than the efficiency gained.

RISK REGISTER

What is most likely to go wrong

RISK	CONSEQUENCE	CONTROL	PLANE
Vehicle misfit or false stock promise	Wasted trip, return, fitting cost, poor review, ACL exposure	Vehicle configuration model, confidence threshold, "call to confirm" for stockists, and no stock claim without a citation	P4
Voice latency blowout	Caller hangs up, containment collapses	Speculative prefetch, two model tiers, latency asserted in CI against recorded turns	P5
Store connectivity loss	Trading stops at the counter	On-premise inference, deterministic fallback, AI off the transaction path	P7
Prompt injection via customer input or catalogue content	Data exfiltration or an unintended action	Per-agent tool allowlists, no secrets in context, output validation, injection classifier	P6
PII to an offshore endpoint	Cross-border disclosure breach	Payload classification and residency routing at the gateway, redaction before egress	P1

RISK	CONSEQUENCE	CONTROL	PLANE
Cost runaway	Gross margin erosion on every automated interaction	Per-conversation budget caps, gateway accounting, alerting on cost per contained call	P5
Over-automation	CSAT and brand damage, hard to walk back	Risk tiering, shadow mode, canary by queue, human-in-the-loop by default	P1
Vendor lock-in on the voice leg	Re-platforming cost at the worst moment	Thin orchestration interface, tool mesh owned internally	P3
Shadow AI use by staff	Ungoverned data leaving the business	Sanctioned tooling, gateway as the only egress, policy with teeth	P1
Transcript retention and erasure gaps	Compliance finding, DSAR failure	Retention schedule, erasure path that reaches the vector index and is tested	P4
Eval drift as the business changes	Silent quality decay after a catalogue or policy change	Datasets refreshed on a cadence, regression alerts, a named owner per agent	P2

OPERATING MODEL

How fifteen people run this

The platform is run as an internal product with a named owner, an SLA and a roadmap, not as a shared library that everyone edits. A platform with no owner becomes a dumping ground within two quarters.

Platform pod · 3–4 Gateway, MCP mesh, retrieval, guardrails, eval harness, observability. Owns the SLAs every channel depends on.	Voice pod · 2–3 Telephony, realtime audio, latency budget, containment quality, compliance of the voice channel.
Commerce & store pod · 2–3 Storefront assistant, POS copilot, offline path, counter workflows.	Data & knowledge · 2 Vehicle configuration model, corpus ownership, data quality, policy rules. Reports to neither channel pod, deliberately.
Platform reliability & security · 2 Egress control, secrets, residency, incident response, the kill-switch runbook.	Product & analysis · 1–2 Use-case intake, risk tiering, before-and-after measurement, the governance forum.

The engineering method is itself a deliverable

The brief asks for someone fluent in AI-first tooling, and that has a concrete architectural consequence: the same discipline applies to the platform's own development. Specifications committed alongside code. Agentic review on every pull request. An internal MCP server for the team's own tooling, so debugging a production conversation is a question rather than a query hunt. Repository-level convention files so every agent working in a codebase inherits the same constraints.

This is not a productivity footnote. A team that uses evaluation gates, versioned prompts and traced failures on its own work finds the same discipline obvious in the platform. A team that does not will ship prompts nobody can audit.

THE FIRST NINETY DAYS, IN ONE LINE

Build the spine, prove it on the staff copilot, and have the vehicle configuration model under way before the voice agent takes its first real call.

Reference architecture · version 1.0 · component names are generic
and map to any equivalent platform. Figures are hand-drawn to scale;
the latency budget reflects overlapped streaming stages, not
sequential summation.